Autologous iPSC-Derived A9 Dopaminergic Progenitors in a Neurotrophin-Loaded Collagen Hydrogel for Parkinson's Disease

Parkinson's disease is a chronic neurodegenerative disorder defined by progressive loss of dopaminergic neurons within the substantia nigra pars compacta (SNpc) and by alpha-synuclein aggregation into Lewy bodies. It ranks among the most common neurodegenerative diseases, and its burden will grow as populations age. Patients develop resting tremor, rigidity, and bradykinesia; over time, postural instability and non-motor problems such as sleep disturbance and cognitive impairment also arise.[1] Current therapies, mainly dopaminergic drugs and deep brain stimulation, relieve symptoms but cannot halt the underlying neuronal loss. This unmet need and the focal pathology make it a compelling target for cell replacement. The neurons predominantly lost are the A9 dopaminergic subtype of the SNpc, which project through the nigrostriatal pathway and deliver dopamine to the striatum, where it modulates basal ganglia circuitry to govern voluntary movement. Because the deficit is confined to one cell type in one region, a neuronal graft is a rational replacement.[2]

A stem cell approach suits this problem because it can provide a virtually unlimited supply of dopaminergic neurons that release dopamine continuously within the striatum, restoring signaling that oral drugs cannot replicate. The proposed graft uses A9 dopaminergic progenitors obtained from induced pluripotent stem cells (iPSCs), reprogrammed from the patient's mature cells, such as skin fibroblasts or peripheral blood cells. This yields an autologous graft and avoids the ethical concerns of embryonic or fetal tissue. Being pluripotent, iPSCs can form any cell type; they are produced by reprogramming with the four Yamanaka factors, OCT4, SOX2, KLF4, and c-MYC, introduced via non-integrating episomal vectors to leave the host genome intact. Where patient-matched cells are impractical, an HLA-matched donor line may be used.[3]

Differentiation recapitulates ventral midbrain development. The iPSCs are first directed to a neural fate by inhibiting SMAD signaling, then patterned with sonic hedgehog and a WNT activator toward a floor-plate identity, the correct origin of authentic nigral neurons.[4] Correctly patterned progenitors express LMX1A, FOXA2, and EN1, predicting a favorable graft. The cells are then sorted using the floor-plate marker CORIN to remove serotonergic neurons, which caused involuntary movements in earlier fetal-tissue trials. The purified progenitors are encapsulated in an injectable collagen hydrogel loaded with the neurotrophic factors GDNF and BDNF. The collagen scaffold supports cell attachment and protects the cells during injection, while gradual release of GDNF and BDNF sustains them through the vulnerable early phase, when most graft cells die.[5] Being injectable, it is delivered into the striatum through a fine cannula in a minimally invasive stereotactic procedure.[6]

This design addresses the three obstacles of earlier grafts: poor cell survival, immune rejection, and contamination by unwanted cell types. In a parkinsonian rat model, progenitors carried within the neurotrophin-loaded hydrogel achieved markedly greater survival and maturation than those given without it, implying that a smaller dose would suffice and the risk of overgrowth would fall.[5] CORIN sorting limits dyskinesia by removing serotonergic cells, and the autologous source reduces the need for immunosuppression and its risks of infection and malignancy; together, these features should improve patient tolerance. If the graft persists, it should restore striatal dopamine, ease motor symptoms, and improve daily function and quality of life. Recent human trials using hESC- and iPSC-derived progenitors have shown graft survival, increased dopamine output on positron emission tomography (PET), and no tumor formation over a two-year follow-up.[6] Although restorative rather than curative, its renewable supply, scalable production, and physiological dopamine delivery make it promising for patients beyond current therapies.

References

1. Kouli A, Torsney KM, Kuan WL. Parkinson's disease: etiology, neuropathology, and pathogenesis. In: Stoker TB, Greenland JC, editors. Parkinson's Disease: Pathogenesis and Clinical Aspects. Brisbane (AU): Codon Publications; 2018. p. 3–26.
2. Stoker TB. Stem cell treatments for Parkinson's disease. In: Stoker TB, Greenland JC, editors. Parkinson's Disease: Pathogenesis and Clinical Aspects. Brisbane (AU): Codon Publications; 2018. p. 161–175.
3. Sawamoto N, Doi D, Nakanishi E, Sawamura M, Kikuchi T, Yamakado H, et al. Phase I/II trial of iPS-cell-derived dopaminergic cells for Parkinson's disease. Nature. 2025;641(8064):971–977.
4. Kriks S, Shim JW, Piao J, Ganat YM, Wakeman DR, Xie Z, et al. Dopamine neurons derived from human ES cells efficiently engraft in animal models of Parkinson's disease. Nature. 2011;480(7378):547–551.
5. Comini G, Kelly R, Jarrin S, Patton T, Narasimhan K, Pandit A, et al. Survival and maturation of human induced pluripotent stem cell-derived dopaminergic progenitors in the parkinsonian rat brain is enhanced by transplantation in a neurotrophin-enriched hydrogel. J Neural Eng. 2024;21(2):024002.
6. Tabar V, Sarva H, Lozano AM, Fasano A, Kalia SK, Yu KKH, et al. Phase I trial of hES cell-derived dopaminergic neurons for Parkinson's disease. Nature. 2025;641(8064):978–983.